

System identification

$n_x = 6$ was read from the singular-value spectrum of the block Hankel matrix (Figure 1): six singular values exceed the relative threshold $10^{-6}\sigma_1$, with the remainder at numerical zero. ERA and CVA+EM were both fitted with $n_x = 6$. CVA+EM showed higher mean absolute error in both magnitude and phase across all $n_u n_y = 32$ input-output channels (Figure 3), and its transfer function failed to reproduce the per-neuron frequency selectivity for input u_1 that ERA recovered (Figure 5). One-step-ahead Kalman prediction on $N = 50$ held-out trials ($T = 2000$ steps) reversed this ordering: CVA+EM achieved a higher median Frobenius R_F^2 (0.399 vs 0.289; Figure 4). ERA was adopted on the basis of its frequency-domain accuracy.

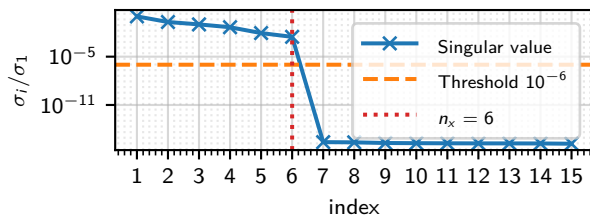


Figure 1: Normalised singular values σ_i/σ_1 of the block Hankel matrix constructed from 55 noise-free Markov parameters (20 block rows, 30 block columns, $n_u = 2$ inputs, $n_y = 16$ outputs). The orange dashed line marks the relative threshold 10^{-6} ; singular values below this level are indistinguishable from numerical zero under exact noise cancellation. The red dotted line indicates $n_x = 6$. [Source](#)

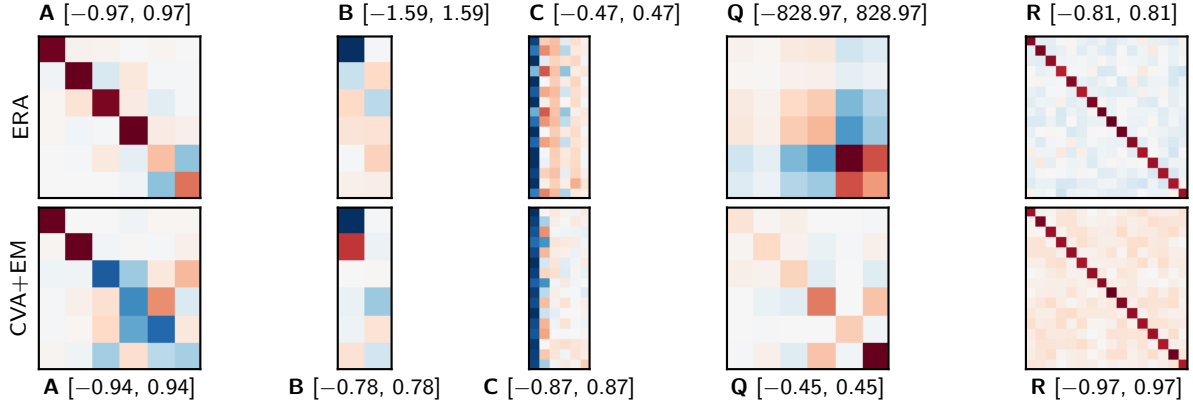


Figure 2: ERA (top row) and CVA+EM (bottom row) estimates of system matrices. Red indicates positive, blue negative; each subplot title shows the symmetric colour range. [Source](#)

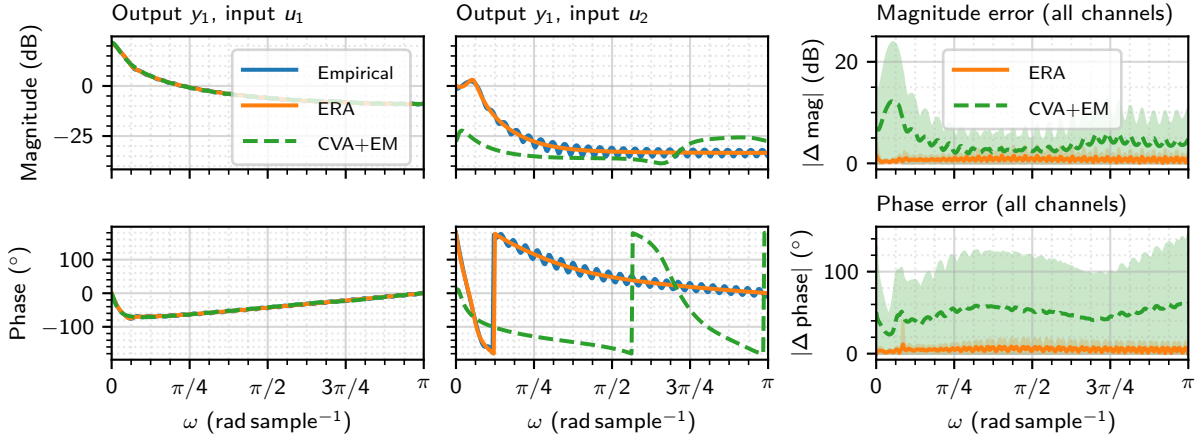


Figure 3: Frequency-response comparison for ERA and CVA+EM identified models. Left and centre columns: Bode magnitude (top) and phase (bottom) for output y_1 driven by inputs u_1 and u_2 respectively, comparing the empirical response (discrete Fourier transform of 55 noise-free Markov parameters, zero-padded to 512 points) against the analytic response $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ for the ERA and CVA+EM models with $n_x = 6$. Right column: absolute magnitude error (top) and absolute phase error (bottom), shown as mean $\pm$ 1 s.d. across all $n_u n_y = 32$ output-input channels for each method. [Source](#)

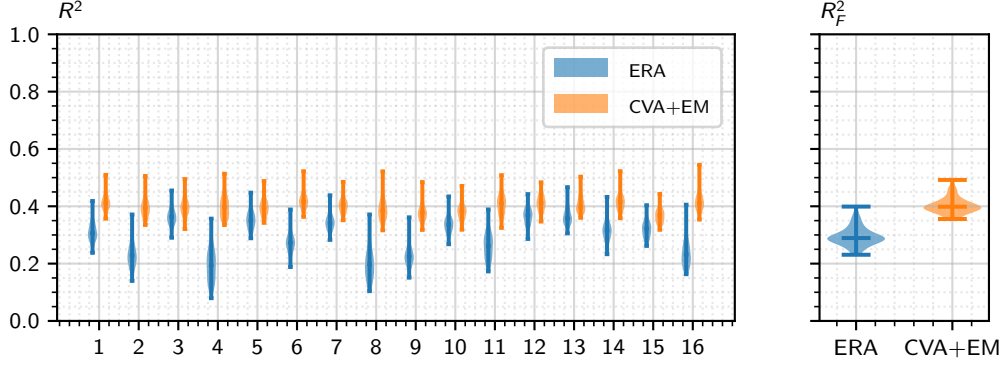


Figure 4: Per-channel R^2 (left, $n_y = 16$) and Frobenius R_F^2 (right) of one-step-ahead Kalman predictions across $N = 50$ held-out trials of $T = 2000$ steps ($n_x = 6$; first 100 steps excluded). ERA median $R_F^2 = 0.289$; CVA+EM median $R_F^2 = 0.399$ (Wilcoxon signed-rank, $p = 1.00e + 00$). [Source](#)

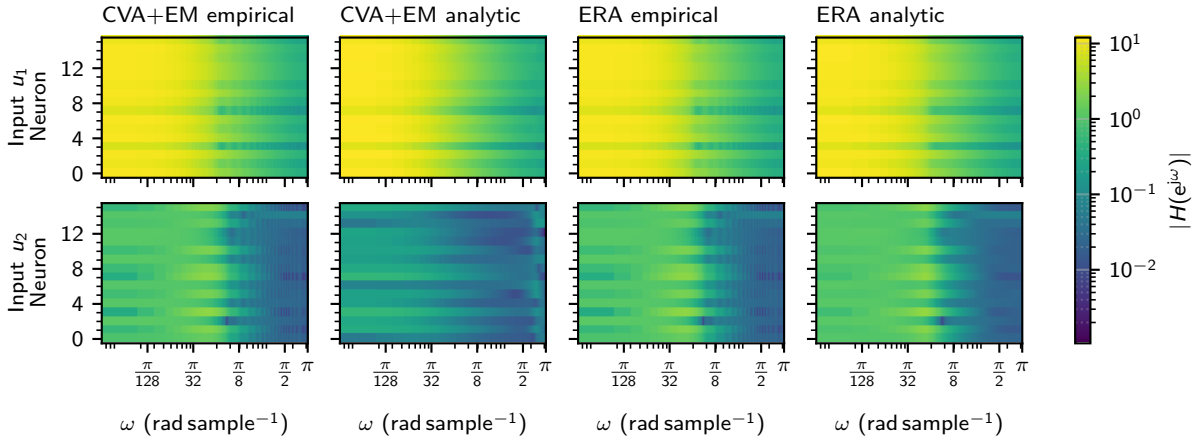


Figure 5: Per-neuron frequency selectivity for inputs u_1 (top row) and u_2 (bottom row). Each panel shows $|H(e^{j\omega})|$ for all $n_y = 16$ neurons (vertical axis) across $\omega \in [0, \pi]$ rad sample $^{-1}$ (horizontal axis), on a shared logarithmic colour scale. CVA+EM (left 2 columns), ERA (right 2 columns); within each pair, the left column shows the empirical response (DFT of 55 Markov parameters, zero-padded to 512 points) and the right column shows the analytic transfer function $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$. [Source](#)